Towards a unified framework for AI integration in the corporate ecosystem

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Abstract

This framework stems from the obsession of someone who, in daily guiding organizations toward AI adoption, has seen the same paradox repeat itself: the abysmal distance between the theoretical elegance of existing models and their concrete implementability. It is an attempt to answer a question that resonates in every meeting room, in every pilot project, in every strategic discussion: how to transform the intrinsic complexity of the AI ecosystem into tangible and sustainable business value? The proposed model articulates the permanent integration of Artificial Intelligence through a two-dimensional structure that combines four levels of integration (from bi-directional data exchange to the definitional integration of the organization) with four interconnected functional domains (Resources, Data and Information, Use Cases, Tools). The architecture distinguishes two fundamental paradigms of human-AI interaction: Co-Intelligence, where AI amplifies human capabilities while maintaining decisional control in the operator, and the Cybernetic Workforce, where human-machine integration reaches levels of sophistication such that the boundary between human and artificial action becomes indistinguishable. This dichotomy materializes through seven types of use cases (Synchronous and Asynchronous Personal Assistants, Enhanced Use, RPA and Intelligent Automation, Analytics, Augmented Employee) implementable through Make-Build/Configure-Buy technological strategies. If organizations truly want to be supported by AI, it is necessary to guarantee democratic access to the technology; for this reason, the framework is completed with the definition of the Corporate AI Portal, a unified digital environment that guarantees access to AI resources through five domains of experience: Use Case Marketplace, Prompt & Code Library, Access to Approved Tools, and the lateral areas of Documentary Resources and Governance. The peculiarity of the model aims to reside in the synthesis between conceptual rigor and operational pragmatism, offering organizations not only a map of the AI ecosystem but a concrete guide to navigate its complexity, transforming the adoption of Artificial Intelligence from episodic experimentation to systemic and permanent transformation. The framework attempts to distinguish itself from existing literature by its ability to overcome the critical fragmentation that characterizes current academic and consulting approaches. While existing frameworks tend to excel in specialist domains—respectively focusing on distributed innovation, development of specific technical capabilities, or structured governance—few offer the systemic integration necessary to address what the most recent research defines as a "crisis of fragmentation" of enterprise AI tools. The present proposal bridges this gap by providing

a two-dimensional architecture that simultaneously orchestrates progressive integration levels with interdependent functional domains, transforming the complexity of AI adoption from an epistemological obstacle into a systematizable competitive advantage.

Contents

1 Introduction (facetious) to the work conducted

Every now and then I venture (I embark, given that I am finalizing this tome in a marine environment during the first days of "summer break" 2025) into an enterprise that inevitably leads me to ask myself: "but why do you do it?". This short (in its initial intentions it was short, I swear) article verbalizes in literal form a series of schemes that I started drawing a few weeks ago during an "intra-meeting" break in a client's meeting room. And as always happens to me, this, which was meant to be a useful exercise for me to understand (or at least try to govern the complexity implicit in understanding), became an obsession that led me to tons of reading, research, writing, and rewriting in order to produce a result that could be shareable as a "status report" or as a "line drawn" (at least at a certain date, given the all too rapid evolution of the subject), on a topic. And as always, this is what I have understood, hypothesized, and theorized up to here. Then we shall see. Some important notes: the schematizations are improvable, I know. Since I know from experience that I never like them and that, in any case, I will redo them every time I have to present them, I accept them as they are... "functional to the objective". To write this article, as always, I availed myself of a couple of "cybernetic collaborators" (Claude and ChatGpt) for research, cross-verification, coherence checking, and post-editing. As always, the contribution was not essential but integrative because it allowed me to notice that some things that were clear to me... were clear only to me. I should extend the use of these "double-blind" techniques to daily life as well; probably everyone who deals with me would benefit from it.

2 Introduction (serious) to the integration model

The advent of Artificial Intelligence as a transformative technology has generated a significant paradox in the contemporary corporate landscape. On one hand, academic and consulting literature has produced an abundance of theoretically sophisticated frameworks for AI integration in the organizational sphere. On the other hand, operational reality reveals a dramatic gap between the conceptual completeness of these models and their actual practical implementation. One of the most recent studies I happened to read highlights that, although 78% of organizations use AI in at least one business function, only 1% describe their implementations as "mature". This data acquires even more marked relevance in the Italian context, where research on SMEs shows AI implementation rates hovering (data from June 2025) around 8%, revealing a particularly pronounced gap towards medium-sized organizations. Surely the time factor plays an important role in this evidence. Any new technology, to be adopted, needs physiological maturation time both from an exquisitely technical point of view (given that every day there is something new, we will have to wait for a minimum plateau to see companies without the fear of making an investment that becomes obsolete the following day) and to find the fit in an organization, whether complex or simple. And this certainly amplifies the need to find models to explain to ourselves what is happening and what will happen. The problem does not lie, paradoxically, in the absence of holistic frameworks because these exist in abundance. The fundamental problem lies in their nature: such frameworks are often located at the extremes of the application spectrum, resulting in either excessively theoretical and abstract for practical implementation, or too specific and sectorial to provide a systemic view of the AI ecosystem integrated into the broader corporate one. This gap is particularly evident when considering the intrinsic complexity of AI as a general-purpose technology, which requires a simultaneous transformation on four critical dimensions: organizational, process, technological, and data. Existing frameworks tend to privilege one or two of these dimensions, leaving organizations, especially medium-sized ones with limited transformation capabilities, without an integrated and pragmatic guide to navigate the complexity of AI adoption. The need to bridge this conceptual and operational gap, which I have felt over the course of this last year, entirely dedicated professionally to guiding organizations toward conscious "Adoption" paths of AI, is heightened by the evolutionary nature (effervescent and disruptive) of the technology and by the growing competitive pressure that organizations face. It is no longer a matter of deciding whether to adopt AI, but how to do so in a systemic, sustainable, and value-generating way. It is in this context that the need emerges for a framework that is at once academically rigorous in its theoretical foundation and practically implementable in its operational application. The present framework is my first attempt to create a comprehensive model for the "permanent" integration of Artificial Intelligence in the corporate sphere, articulated through a multidimensional structure that embraces four levels of integration and four interconnected functional domains. Its peculiarity lies in the search for balance between analytical depth and practical applicability, designed specifically to respond to the needs of organizations that need a unitary and implementable view of the AI ecosystem. The definition of a corporate AI ecosystem represents an epistemological and practical challenge of notable complexity. Unlike previous technologies, characterized by relatively linear and discrete implementations, AI manifests systemic properties that transcend traditional boundaries between technology, organization, processes, and data. This intrinsically interconnected nature generates what we could define as an "eco-systemic effect", where the impact and effectiveness of AI emerge from the quality of relationships and interdependencies rather than from individual technological components. Existing literature, while recognizing this complexity, has largely failed to provide adequate operational tools. The most sophisticated academic frameworks demonstrate excellent dimensional coverage from a theoretical point of view but are lacking in the cultural and contextual adaptation mechanisms necessary for practical implementation. In parallel, consolidated consulting methodologies, while offering detailed operational guidance, tend to privilege approaches that often do not take into account the variety and specificity of organizational contexts. This conceptual and operational fragmentation assumes particular relevance in the context of medium-sized organizations, which represent the prevalent productive fabric in many advanced economies and in particular the Italian one. Such organizations find themselves in a paradoxical position: they possess the organizational flexibility necessary for innovative AI implementations, but often lack the resources and specialist skills to navigate the complexity of existing frameworks. The result is a widespread underutilization of the transformative potential of AI, with implementations that remain confined to isolated use cases rather than evolving toward integrated ecosystems.

3 Architecture of the framework: (attempted) synthesis between rigor and pragmatism

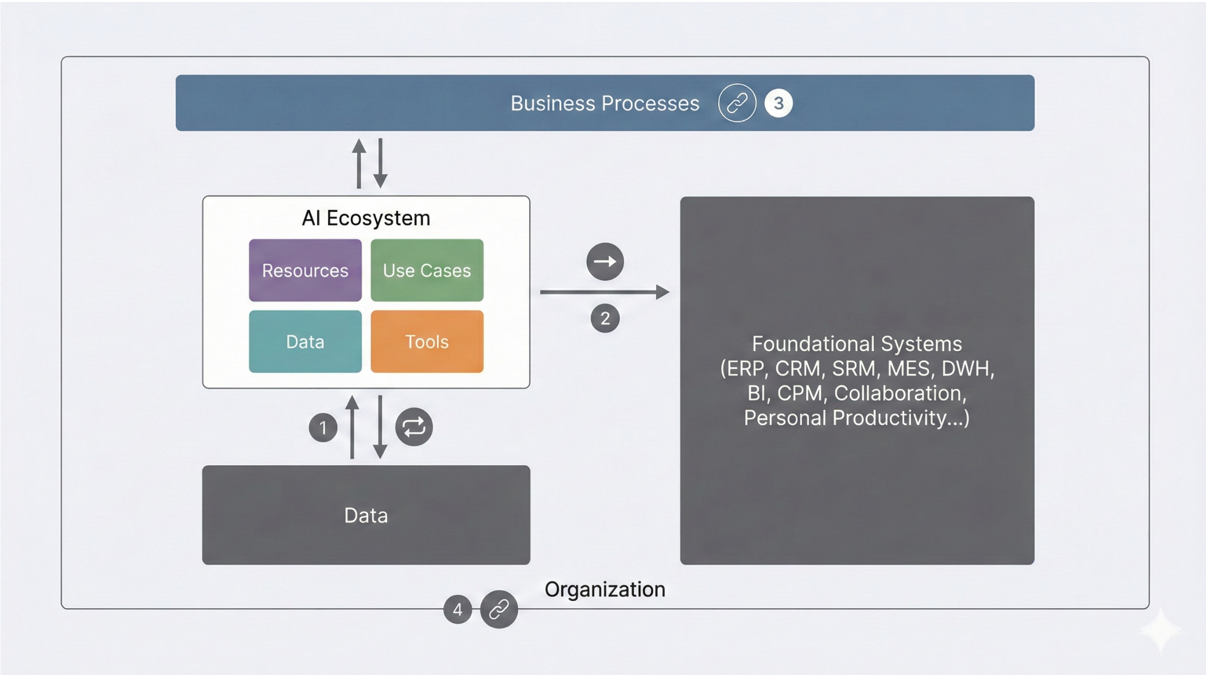

Figure 1: The AI ecosystem and its interconnections with the corporate ecosystem

The need for a new conceptual approach emerges from the awareness that AI integration requires a simultaneously systemic and pragmatic vision. The framework proposed here responds to this need through a two-dimensional architecture that combines four levels of integration with four interdependent functional domains, thus offering a complete but operationally manageable conceptual map (hopefully, at least).

3.1 The four levels of integration: from coexistence to transformation

The framework identifies four evolutionary modes of AI integration, implicitly stating that, to be effective and efficient, AI must be grafted into the corporate context, each characterized by specific interaction dynamics and increasing levels of organizational transformation:

Level 1 - Bi-directional Data Exchange The AI ecosystem operates as an interconnected system that withdraws, uses, organizes, and consumes data from the corporate context, simultaneously generating new data that is returned and integrated into the system itself. This represents the fundamental level of interaction, characterized by a dynamic and circular flow of information. However, it is fundamental to remark that unlike other technological clusters which, from a data point of view, are containers to be filled, in the case of Generative AI, the models, even when just adopted, have their own "initial" dowry (that deriving from training) which brings into the company implicitly a large amount of data and information (which can also be carriers of "bias") that immediately enter into a relationship with corporate data.

Level 2 - Unidirectional Action AI tools act directly on foundational systems through operations of access, actuation, piloting, and command. This level represents a more operational and direct interaction, where AI can assume an active role in the management of existing systems with a view to automation or "agency".

Level 3 - Definitional Integration of Processes The AI ecosystem integrates into corporate processes, defining proprietary processes and altering existing ones. This level entails a structural transformation of organizational workflows, where AI becomes an integral part of operational logic.

Level 4 - Definitional Integration of the Organization The deepest level of integration, where the AI ecosystem inserts itself into the organization defining new organizational structures, roles, and competencies of its own, while modifying existing ones. It represents the most important transformation of the corporate setup.

3.2 The four domains of the ecosystem: interdependence and systemic complementarity

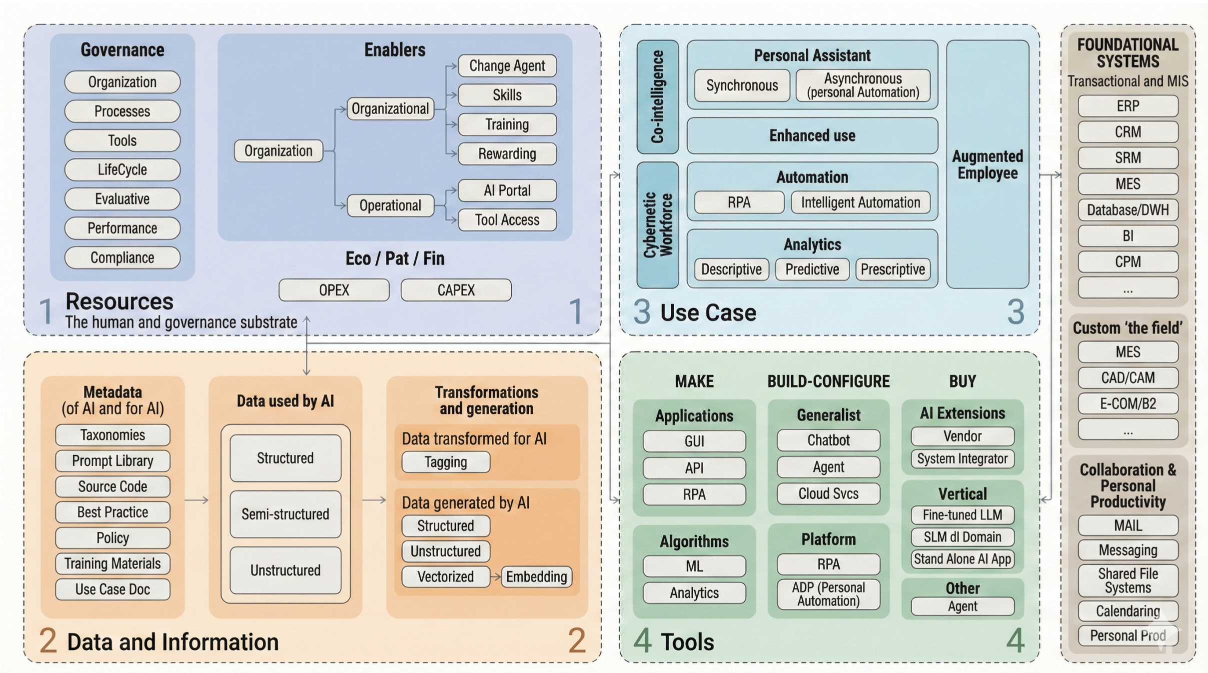

Figure 2: The four macro domains of the ecosystem

The second dimension of the framework articulates the AI ecosystem through four functional domains that operate in systemic synergy. Unlike most existing frameworks, which tend to treat these domains as separate components to be optimized individually, the present approach emphasizes their intrinsically interconnected nature and the need for integrated design.

Resources The set of all "soft" resources mainly focused on HUMAN resources that sustain, support, and constitute the backbone of the ecosystem.

Data and Information The domain of AI data and information with the declination between context data, those used (consumed), transformed, and generated by AI.

Use Case The articulation of use cases by typology and nature. The estimate to date sees a balanced distribution between Co-Intelligence and Cybernetic Workforce but in the latter a prevalence still of automation.

Tools The variety of available tools suggests the measure of complexity. Each nature of use case can be articulated on one (or sometimes more than one) different tool.

3.3 The need for a systemic vision

The convergence of these four domains through the four levels of integration generates a systemic complexity that transcends the sum of the individual parts. It is in this complexity that resides both the main challenge of AI adoption and its deepest transformative potential. However, this systemic complexity requires adequate conceptual and operational tools. It is here that most existing frameworks demonstrate their limits: excellent in the description of individual components, but insufficient in guidance for integrated orchestration. The present framework responds to this gap by providing a conceptual map that is simultaneously complete in its systemic vision and pragmatic in its operational applicability, designed specifically for the needs of organizations that need to navigate AI complexity without having unlimited resources for experimentation.

4 Detailed exploration of the four framework domain areas

Before proceeding with the detailed exploration, it is fundamental to contextualize the need for this unified approach in light of the systemic fragmentation that characterizes the current landscape. Recent studies highlight how organizations implement on average dozens of different AI tools even in single functional domains, while a significant percentage of initiatives remains stuck in what is defined as "pilot purgatory" - the inability to move past the experimental phase due to the impossibility of unified orchestration. This fragmentation crisis stems precisely from the absence of frameworks that simultaneously address all four dimensions of the AI ecosystem: organizations excel in the selection of sophisticated tools but struggle in the systemic integration that transforms isolated pilots into sustainable transformation. The present framework responds to this critical gap by providing the conceptual and operational architecture necessary to overcome the governance-implementation disconnection that characterizes the majority of current corporate AI initiatives.

4.1 Domain "Resources"

The resources from which the ecosystem benefits represent, in large part, components that pertain prevalently to soft aspects and in particular aspects related to the HUMAN actors who intervene and interact with other parts. We can identify Governance resources, resources that act as Enablers in the adoption and use process, and resources generically of a Financial type.

4.1.1 Governance resources

Governance resources consist of organizational, process, and tool elements governing them. Organizational elements are those that enter into a relationship with the corporate ecosystem in the ways we introduced previously. To govern the introduction and effective and efficient use of AI in the company, dedicated organizational structures are certainly needed (which must therefore be introduced or "derived" from existing ones) such as, for example, a "strategic" committee that defines objectives, methods, and purposes of the technological transformation, a center of "excellence" that oversees the adoption process and manages the projects that are identified from time to time, also defining aspects related to technologies and organizational impacts, and an operational committee that defines priorities and involvement of the different teams participating in the program. The governance of these structures is implemented through dedicated processes that interact with the ecosystem of corporate processes defining new processes and altering others. The new processes, operated by the organizational structures identified above, mainly concern the management of the use case lifecycle (from identification to qualification to realization to operation) and evaluative ones both in the ambit of performance and in that of compliance (i.e., in respect and adherence to the various regulations in place - first and foremost the EU AI Act). The introduction of AI, however, also transforms existing processes, not only IT ones that must be "integrated" in order to foresee new cases and new "technological natures" to identify, evaluate, acquire, and manage, but also those relating at least to the Legal and HR worlds. In the domain of legal support to the company, new and specific skills related to risk assessment and safeguard procedures to be adopted must be developed and regulated in specific processes or activities. In the human resources domain, "core" processes (selection, training, talent management) must be adapted to include content themes relating to AI (to remove HR from the embarrassment of not knowing how to answer questions that are now recurrent like "what is the company's policy towards AI?" but also to equip them with an adequate set of questions (and answers) to administer to candidates in order to evaluate their competence in this regard). In order to be able to efficiently manage new or transformed processes, the defined governance structures must equip themselves with tools. Due to the nature and typology of the processes we are considering, the tools we speak of are performance management and evaluation tools (such as dedicated KPIs on adoption, impact in terms of savings or effectiveness, reporting, dedicated business cases, etc.) or compliance tools (evaluative models, registers in which to report considerations and conclusions relating to riskiness according to current criteria, etc.). The implementation of a corporate AI ecosystem introduces risk categories that require specific management approaches integrated into the general framework. In addition to traditional cybersecurity and compliance risks, AI generates systemic risks linked to algorithmic bias, model drift, technological dependence, and unforeseen organizational impacts. These risk management mechanisms must be embedded natively in every use case and integration level, transforming compliance from an external constraint into a competitive advantage through greater reliability, transparency, and social acceptance of the implemented AI solutions.

4.1.2 Resources acting as transformation enablers

To operationalize transformation efforts effectively and efficiently, resources acting as enablers are needed, i.e., as transmission belts to transform theoretical intents into tangible results. A first group of enabling resources are certainly, once again, those of an organizational type. To make the transformation widespread and substantial, a network of change agents is needed, i.e., resources who are guaranteed adequate training, on whom an investment is made for the development or strengthening of skills suitable to act as agents for collection, evaluation, and possibly response (in relation to the nature of the use cases) to the needs of the structures to which they belong. In order to make the commitment of these resources sustainable and permanent, it is necessary to provide specific rewarding that recognizes and rewards the results achieved. To channel energies and not disperse efforts, operational tools are needed that are tendentially simple to implement but often neglected in their relevance: first of all, widespread and "democratic" access to tools must be guaranteed to the entire organization. Clearly, behind the term "ENTIRE" hide a series of details that are crucial in the adoption and use path/process and that then reverberate on the ecosystem (which by its nature represents a situation "in evolution but established" and not a project-type vision). Preliminary is clearly the decision by the Center of Excellence of which technology and technologies, which tool and tools, as well as the definition by the Strategic and Operational Committee of the organizational scope of the transformation, of priorities, and of "waves" of adoption. But once these aspects are clarified, the plan is in execution, and adoption is proceeding, it becomes fundamental to remove what still remains a stigma (the use in the corporate environment of generative AI tools today, precisely because it is not "managed" and governed in most cases, is more so than one believes) by giving extensive access to tools to all potential users. Access that must be governed in a structured way through an "AI PORTAL" that collects, organizes, and makes accessible tangible resources in the form of data and metadata (see below), use cases, and tools.

4.1.3 Resources of "financial" nature

However improper (reductive mostly) the reference to the "financial" nature (also because in this block of resources, agreements and contracts with strategic suppliers etc. would essentially fall), it is important not to forget that the AI ecosystem (and the AI adoption process in general) entails investments (one-shot capex for adoption projects, for training, for mobilizing people, for the development of eventual tools) and operating costs (licenses, spaces, and computing capacities acquired on cloud platforms, as well as the cost of "human" resources for the share of time they dedicate to transformation). The economic evaluation of the AI ecosystem requires a sophisticated approach that goes beyond traditional CAPEX/OPEX models, incorporating metrics specific to the evolutionary and systemic nature of artificial intelligence. Traditional ROI models prove inadequate for technologies that generate value through network effects, continuous learning, and process transformation rather than simple automation. A complete economic framework for AI must include KPIs and specific indicators that must be integrated into executive dashboards providing real-time visibility on the evolution of AI return on investment, enabling informed strategic decisions on resource allocation and the acceleration or modification of implementation priorities.

4.2 Domain "Data and information"

The domain of data and information constitutes the fundamental substrate on which the entire AI ecosystem rests, representing not only the fuel that powers artificial intelligence systems but also the result produced by their processing activity. The peculiar nature of AI, which consumes data to produce new data and information, generates a complex circular flow that requires articulated and multidimensional management. We can identify four macro-categories of data that interact dynamically in the ecosystem: Metadata (of AI and for AI), Data used by AI, Data transformed for AI, and Data generated by AI.

4.2.1 Metadata (of AI and for AI)

Metadata represent the informational backbone that supports and governs the entire AI ecosystem, acting as a connecting element between the different components and ensuring the coherence and governability of the whole. This category comprises fundamental elements such as taxonomies, which provide the classificatory structure necessary to systematically organize and categorize all elements of the ecosystem (use cases, tools, data typologies, skills, etc.) and not only (although primarily) the data and documents accessed by AI, guaranteeing a common language and a shared vision within the organization.

The Prompt Library constitutes a strategic repository that collects, organizes, and makes accessible interrogation and instruction models optimized for the AI tools in use, representing a procedural knowledge asset that grows and evolves with the organization's usage experience. The Source Code includes all source code developed internally for customizations, integrations, and personalized solutions that characterize the specific implementation of the ecosystem in the organization. Best Practices enclose the experiential knowledge accumulated in the adoption and use of AI, constituting a wealth of methodologies, approaches, and consolidated solutions that guide operational and strategic decisions as well as the "instructions for use" released by solution providers.

Policies define the internal normative framework regulating the use of AI, establishing principles, limitations, responsibilities, and security procedures. Training Materials organize structured educational content for the development of AI skills at all organizational levels. Finally, Use Case Documentation systematizes the description, configuration, and results of all use cases implemented or being implemented, constituting a fundamental application knowledge repository for the replicability and scalability of interventions.

4.2.2 Data used by AI

This category represents the informational universe that constitutes the primary input for AI systems and is articulated according to different structuring hierarchies. **Structured data** comprise all information organized according to predefined schemas and formal relations, typically residing in corporate databases, ERP systems, CRM, DWH, and BI and generally in all transactional systems that manage information according to consolidated entity-relationship or (multi) dimensional models. These data benefit from high intrinsic quality due to integrity and consistency controls implemented in source systems, but often require harmonization and normalization processes to be effectively used in AI contexts requiring integrated and transversal views compared to original functional silos. **Semi-structured data** include information possessing some elements of organization (such as accounting or management documents, XML, JSON files, log files, emails with metadata) but which do not adhere to rigid predefined schemas. This typology requires parsing and standardization processes to extract informative value usable by AI systems. **Unstructured data** represent the most complex and volumetrically significant category, comprising free text documents, images, videos, audio, presentations, and all corporate information assets not intrinsically organized according to formal schemas. The valorization of this category requires advanced natural language processing, computer vision, and machine learning techniques to extract meaning and structure from intrinsically unstructured content.

4.2.3 Data transformed for AI

The passage from "raw" data usable by AI to data actually processable by systems requires transformation processes that prepare information for optimal use. Tagging represents the annotation and labeling process that enriches data with meta-information facilitating its categorization, search, and contextual use. This process can be automated through machine learning algorithms or require human intervention to guarantee accuracy and relevance of annotations. *Embedding* constitutes a fundamental transformation that converts information of different natures (text, images, audio) into numerical vector representations maintaining original semantic relationships in a multidimensional mathematical space. This transformation is essential to allow AI systems to process and compare information of different natures using standardized mathematical operations. The embedding process, besides numerical conversion, entails the creation of vectorized representations constituting the final form in which information is actually processed by AI models. The quality and dimensionality of these representations directly influence the performance and accuracy of the AI systems utilizing them.

4.2.4 Data generated by AI

The output of the processing activity of AI systems generates new categories of data enriching the corporate information ecosystem. Generated structured data include analysis results, numerical forecasts, classifications, scoring, and quantitative insights produced by AI systems which can be immediately integrated into corporate decision-making processes or existing information systems. These data maintain structuring characteristics facilitating their immediate operational use. Generated unstructured data comprise textual content, images, audio or video produced by generative systems, synthetic documents, summaries, translations, and all forms of unstructured content emerging from the creative and processing activity of AI. A particular and strategically relevant category is constituted by generated vectorized data, i.e., mathematical representations that AI systems create in their internal processing process and which can be stored and reused to improve future performance, to facilitate similarity search operations, or to form the basis of specialized vector knowledge bases. All data generated by AI must be managed with particular attention to traceability, quality, and governability, considering that they often constitute input for critical decision-making processes and that their reliability depends on the quality of input data and the robustness of the models used.

4.2.5 Integration and information flows

The AI data ecosystem is not constituted by static categories but by dynamic flows and continuous interactions between different typologies. Metadata guide and govern the use of data used by AI, transformation processes are informed by best practices and policies contained in metadata, data generated by AI can become input for subsequent processing or enrich the metadata asset with new taxonomies and classifications. This circular and self-feeding nature requires sophisticated information architectures ensuring consistency, traceability, and quality of the entire information flow, from source data collection to the valorization of generated data.

4.3 Domain "Use case"

The articulation of use cases constitutes the most relevant "operational" dimension of the AI ecosystem, representing the concrete ways through which artificial intelligence technologies integrate into work and organizational processes to generate tangible value and measurable transformation. The classification is based on the fundamental distinction between two human-machine interaction paradigms reflecting different philosophies of collaboration, increasing degrees of technological integration, and differentiated levels of automation of cognitive and operational processes. The cited paradigms can also be considered "usage models" since the term USE CASE in an evolving context like the one we are in cannot refer uniquely (and erroneously) to Task automation but must necessarily consider progressive scopes of activity transformation (or consider redefinitions or new activities entirely). On one hand, we find **Co-Intelligence**, which according to Ethan Mollick's definition represents a "centaur" use of technologies where human intelligence and artificial intelligence collaborate maintaining distinct roles, clearly defined responsibilities, and a recognizable separation between human and machine contribution in the decision-making process. On the other hand lies the **Cybernetic Workforce**, resembling the "cyborg" paradigm where human-machine integration reaches a level of sophistication such that it becomes progressively difficult to distinguish who does what, creating a hybrid, fluid, and self-adaptive workflow where human decisions and intelligent automations merge into an operational continuum. The current estimate based on empirical observations and adoption trends sees a relatively balanced distribution between Co-Intelligence and Cybernetic Workforce in terms of the number of implementations (although, trending, the expectation is for a greater skew in favor of the former). Within the Cybernetic Workforce, there is still a significant prevalence of traditional automation use cases compared to more sophisticated and strategically differentiating approaches like the Augmented Employee.

4.3.1 Co-Intelligence

Co-Intelligence represents the category of use cases where AI acts as an intelligent collaborative partner to the human operator, maintaining a clear and functional distinction between specific skills, operational responsibilities, and domains of expertise of each party in the work process. This mode is characterized by the deliberate preservation of final decision-making control by the human user, who uses AI as a tool for amplifying their cognitive capabilities, accelerating analysis processes, and improving output quality, without ever completely delegating the decision-making process to the machine or losing critical supervision over the results produced. The Co-Intelligence paradigm is founded on the complementarity between human intuition and AI computational capacity, between human creativity and AI generative capabilities, between human contextual judgment and AI pattern recognition, creating synergies that exceed the performance obtainable by individual components operating in isolation. The prevalent purpose is that of "individual super-productivity" intended as individual and general-purpose support for reasoning and retrieving unknown information and the possibility of delegating tasks that could be done alone to recover efficiency.

Personal assistant - synchronous This subcategory comprises all direct, immediate, and real-time interactions with generative AI tools through structured prompts, guided conversations, and collaborative work sessions on generalist platforms requiring the active and continuous presence of the user. Typical and consolidated examples include intensive work sessions with ChatGPT, Claude, Gemini, Copilot, or other conversational platforms where the user formulates specific, detailed, and contextualized requests to receive immediate responses used to complete complex cognitive tasks. These tasks range from writing technical or creative documents to in-depth analysis of multidimensional business problems, from generating innovative ideas for projects or campaigns to resolving technical queries requiring specialist skills, from accurate translation of content maintaining cultural and sectorial nuances to intelligent synthesis of long documents extracting key insights, from creating creative content respecting brand guidelines and communication objectives to elaborating problem-solving strategies for complex operational challenges, up to generating creative alternatives for strategic decision-making. The interaction is intrinsically synchronous as it requires the active, continuous, and cognitively engaged presence of the user who dynamically guides the dialogue, critically evaluates received responses, provides contextual feedback to improve accuracy and relevance, iteratively reformulates requests to obtain progressively more refined results adhering to specific needs, and maintains strategic control over the entire process to ensure alignment with objectives and quality standards. The quality and utility of the final result depend significantly on the prompt (and today more than ever context) engineering skills of the user, on their capacity for structured thought and articulation of requirements, on their ability to iterate effectively with the system through conversation management techniques, and on their competence in providing adequate context, specific constraints, and constructive feedback guiding AI towards usable, accurate, and strategically aligned outputs. This class is strongly dynamic in its evolutions because it obviously corresponds to consumer usage of the most widespread tools (therefore consider as parts of this usage model also "implicit" RAGs such as Claude projects or ChatGPT agents).

Personal assistant - asynchronous This category includes the strategic use of configurable AI agents activated via generalist tools that can operate with a significant degree of temporal and operational autonomy, releasing the user from the need for continuous supervision, constant presence, and real-time intervention during the execution of predefined tasks. Representative and high-impact examples are highly configurable agents available on platforms like OpenAI's custom GPTs (allowing personalization through specific instructions, specific knowledge bases, and behavioral instruction), Claude's customizable agents (trainable on specific tasks and domain skills), or those activated through MCP (Model Context Protocol) connectors allowing fluid integration with external data sources, third-party APIs, and diverse corporate systems to create complete automation workflows. Strategically falling into this category are also all sophisticated personal automations realizable with personal automation tools like Microsoft PowerPlatform (combining AI capabilities with process automation, workflow orchestration, and system integration), or advanced agents developable with Copilot Studio which can be programmed to execute complex multi-phase workflows combining intelligent interactions with multiple applications, data sources, and automated decision points. The distinctive and strategically precious characteristic is the possibility to confidently delegate structured but cognitively demanding tasks executed by AI with limited supervision, programmed control, and exception-based intervention. These tasks range from systematic and intelligent monitoring of specific information sources with automatic filtering based on relevance criteria to periodic elaboration of complex reports based on predefined templates incorporating data analysis, trend identification, and insight generation. They also include automatic management of approval workflows following sophisticated business rules with escalation logic and stakeholder notification, execution of data processing operations according to established parameters including validation, transformation, and automatic quality control, automatic generation of personalized content based on time triggers or specific company events, proactive monitoring of competitive intelligence sources with automatic alerts on significant deviations from established baselines and strategic thresholds, and execution of research tasks combining multiple information sources with synthesis and summary generation maintaining accuracy and completeness standards. Strategic asynchronicity allows the user to set up the agent with detailed instructions, complete context, and clear success criteria, and receive high-quality results at a later optimal time, freeing precious cognitive time for high value-added activities, strategic thinking, and value tasks for the human while AI effectively manages repetitive tasks, time-consuming tasks, or tasks requiring extensive information processing needing intelligence and contextual understanding superior to simple automation.

Enhanced use Enhanced Use represents the natural and strategically solid evolution of traditional corporate systems through the native integration of AI functionalities provided directly by the system vendors themselves or implemented by specialized system integrators with deep domain competence. This category is characterized by the strategic use of preconfigured, carefully tested, and enterprise-ready AI extensions and modules that significantly enrich the existing functionalities of consolidated corporate systems without requiring complex custom developments, significant architectural redesign, or technological integrations introducing instability or security vulnerabilities. The Enhanced Use approach distinguishes itself by minimizing interruptions to existing workflows while maximizing value delivery through the improvement of capabilities that are native to the system and therefore automatically benefit from existing security frameworks, user access controls, data governance policies, and already consolidated operating procedures. Typical, widely adopted, and strategically impactful examples include the full use of AI modules sophisticatedly integrated into enterprise ERP systems like SAP HANA (with advanced machine learning functionalities for demand planning optimization, inventory management intelligence, and supply chain forecasts leveraging historical transaction data, external market signals, and company seasonality patterns), Oracle AI applications strategically incorporated into Oracle Cloud Applications for improving sales forecast accuracy, customer churn prediction with early warning systems, and financial planning automation utilizing complete customer data, transaction history, and market intelligence. Other significant examples are mission-critical CRM systems like Salesforce Einstein completely integrated for lead scoring optimization, sales opportunity prediction with probability assessment, and automatic email marketing personalization improving sales productivity and customer engagement effectiveness, Microsoft Dynamics 365 AI functionalities distributed for generating customer insights, optimizing sales performance through predictive analytics, and supply chain management intelligence providing actionable recommendations based on integrated corporate data and market analysis, in engineering-focused CAD systems like Autodesk AI for design optimization automation, generative design capabilities, and simulation improvement, intelligent SolidWorks functionalities for automatic simulation, test optimization, and design validation accelerating product development cycles, or in mission-critical collaboration platforms like Microsoft 365 Copilot integrated seamlessly for meeting synthesis automation, document analysis improvement, and content generation assistance improving knowledge worker productivity while maintaining corporate security and compliance standards. The Enhanced Use approach is strategically distinguished by ease of adoption due to native integration significantly eliminating the learning curve, reliability deriving from complete vendor testing and quality assurance, and simplified governance automatically inheriting policies, controls, and procedures already implemented for the main system. At the same time, it presents limitations in terms of customization flexibility compared to ad hoc developed solutions and dependence on vendor roadmaps for future improvements and capability evolution.

4.3.2 Cybernetic workforce

The Cybernetic Workforce represents the most advanced and strategically transformative paradigm of integration between human and artificial capabilities, where the traditional boundary between human action and machine action will become progressively indistinguishable, blurred, and functionally irrelevant in the organization's daily operational flow. In this revolutionary category, AI is not limited to assisting the human operator by providing suggestions, analyses, or recommendations, but becomes an integral, active, and decisional part of the work process itself, creating a sophisticated hybrid system that fluidly combines human judgment, artificial intelligence automation, and adaptive learning capabilities in a fluid, dynamic way, contextually responsive to variable operating conditions and evolving business requirements. The distinctive and transformative characteristic of this paradigm is the creation of hybrid workflows where operational responsibility, execution authority, and decision-making process are distributed between human and artificial intelligence in a fluid, contextual, and often indistinguishable way for external observers or even for the operators involved themselves, creating workflow efficiency and capability improvement significantly exceeding performance obtainable by human-only or AI-only approaches. The prevalent purpose is that of process productivity (seen with a corporate perspective) in terms of automation of repetitive corporate processes with a necessary cognitive layer and internal and/or external support in providing information or making decisions in a specific ambit.

Automation Automation constitutes the fundamental and procedural pillar of the Cybernetic Workforce and currently represents the most mature, widely adopted, and empirically proven category within this revolutionary paradigm, articulating itself in two main subcategories representing increasing levels of technological sophistication, integration complexity, and strategic value for the organization. This dimension strategically focuses on the systematic transformation of manual processes, which are time-consuming and error-prone, into intelligent automated flows freeing precious human resources for higher value-added activities, strategic thinking, and properly human tasks requiring creativity, empathy, and complex judgment, while simultaneously improving accuracy, consistency, speed, and cost-effectiveness of mission-critical operational processes.

RPA (Robotic Process Automation) RPA represents the complete automation of structured, rule-based, and highly repetitive processes through mature enterprise platforms like UiPath, Automation Anywhere, Blue Prism, or other consolidated solutions that have demonstrated reliability, scalability, and corporate readiness through thousands of successful implementations in diverse sectors. These powerful tools allow systematically automating operational tasks involving complex interactions with multiple digital interfaces, accurate manipulation of structured and semi-structured data, reliable transfer of information between different systems that do not communicate natively, precise execution of predefined procedural workflows, and high-volume management of high-frequency processes with deterministic logics guaranteeing consistency and accuracy. Typical, widely implemented, and high-ROI examples include complete automation of data entry processes from structured documents to enterprise management systems with automatic validation and error control, automatic management of complex administrative practices following predefined paths like expense approvals, purchase order processing, or customer onboarding management requiring coordination between departments and multiple systems, automatic processing of invoices and orders with extensive consistency checks, validation against contracts and master data, and fluid integration with financial systems for accounting automation, scheduled and reliable synchronization of data between different systems utilizing incompatible formats or different protocols to maintain data consistency across corporate architecture, automatic execution of extensive compliance checks based on complex business rules, regulatory requirements, and internal policies with automatic generation of audit trails and exception reports, automatic generation of complete standard reports through effective aggregation of data from diverse multiple sources with consistent formatting and automatic distribution to appropriate stakeholders, or automatic management of backup procedures, disaster recovery workflows, and system maintenance tasks following consolidated operational playbooks to ensure business continuity and system reliability. Traditional RPA operates according to well-defined deterministic logics and requires well-documented, standardized processes with limited variability to reach optimal effectiveness, but offers high reliability, measurable ROI in the short term, and significant risk mitigation through error reduction and process consistency improvement.

Intelligent Automation Intelligent Automation represents the strategically significant evolution of traditional RPA through sophisticated integration with cutting-edge AI technologies, particularly state-of-the-art Large Language Models and advanced machine learning tools, to successfully manage complex operational situations requiring contextual understanding capabilities, accurate interpretation of unstructured content, sophisticated reasoning, and adaptive decision-making based on logics learned through experience rather than simply programmed through deterministic rules. This evolution represents a quantum leap in automation capabilities allowing management of variable scenarios, intelligent exception management, and adaptive responses to changing business conditions. Characteristic and strategically impactful examples include sophisticated document understanding for accurate extraction and interpretation of critical information from complex unstructured documents like multi-page contracts, variable format invoices, context-rich email communications, or regulatory documents requiring deep understanding of legal terminology and business implications, intelligent processing of emails and complete customer communications with automatic classification based on content analysis, sophisticated sentiment analysis for emotional state assessment and intelligent routing to appropriate teams based on skill matching and workload optimization, automatic classification of diverse corporate content based on machine learning algorithms trained on organizational taxonomy for intelligent categorization, automatic tagging, and strategic archiving supporting knowledge management and regulatory compliance, automatic sentiment analysis and emotion detection on diverse customer feedback for intelligent prioritization and automatic escalation based on severity assessment and customer value analysis, automatic management of exceptions and non-standard cases in automated processes through dynamic decision trees adapting based on historical results and continuous learning from resolution patterns, sophisticated automatic reasoning for complex problem solving and problem resolution in challenging IT contexts or operational environments requiring technical skills and systematic diagnostic approaches, or intelligent management of adaptive workflows modifying dynamically based on current operating conditions, historical performance metrics, and continuous optimization algorithms to maintain peak efficiency while adapting to changing business requirements. Intelligent Automation possesses the ability to adapt to variations in input data, understand important contextual nuances, learn from historical patterns to improve future performance, and make decisions based on sophisticated probabilistic logics rather than just programmed deterministic rules, representing a significant paradigm shift compared to traditional RPA in terms of flexibility, adaptability, resilience, and ability to manage increasing complexity while maintaining appropriate reliability and governance. Wishing to introduce a further differentiation, we could cite autonomous agents here as a cogent model evolution. In this sense, we can consider intelligent automation as an automation following a process with the capacity to clear difficult tasks or manage exceptions, while the autonomous agent as a "computer organism" pursuing an objective based on a set of skills, i.e., the ability to search for or create useful information to complete the task.

Analytics Analytics represents the strategic and intellectually demanding domain of intelligent processing of corporate data for generating actionable insights, accurate and reliable forecasts, and strategic recommendations supporting informed decision-making at all organizational levels from tactical execution to strategic planning. This sophisticated category exploits cutting-edge machine learning algorithms, rigorously rigorous statistical modeling, and advanced AI techniques to transform the vast heritage of corporate data accumulated over time into concrete informational value, actionable competitive intelligence, and measurable strategic guidance, articulating itself in three levels of sequentially increasing technological sophistication, mathematical complexity, and strategic added value for corporate performance. The progression through these levels reflects the evolution of the organization's analytical maturity, its ability to exploit data assets for sustainable competitive advantage, and the sophistication of decision-making capabilities it can achieve through systematic data-based approaches.

Descriptive Analytics Descriptive Analytics focuses on the systematic and complete analysis of extensive historical data to deeply understand what happened in the organizational past and accurately identify underlying causes of observed phenomena, utilizing mathematically sophisticated data mining techniques, rigorous multivariate statistical analysis, and advanced data visualization to identify significant patterns, emerging trends, hidden correlations, and anomalies that can inform strategic decisions and operational improvements. This analytical base is essential for organizational learning and provides the necessary context for understanding complex business dynamics. Consolidated, strategically important, and high-impact examples include complete sales performance analysis for precise identification of best-performing products, most effective channels, most profitable customer segments, and most promising geographic territories with extensive drill-down capabilities to understand underlying success drivers and replicate best practices in the organization, sophisticated customer behavioral analysis through advanced web analytics, detailed customer journey mapping tracking every interaction touchpoint, and mathematical cohort analysis to understand engagement patterns, conversion funnels, retention drivers, and satisfaction factors influencing customer lifetime value, complete operational analysis for systematic identification of process bottlenecks, hidden operational inefficiencies, and significant optimization opportunities through advanced process mining and detailed workflow analysis revealing waste areas and substantial improvement opportunities, creation of sophisticated dynamic and interactive dashboards for real-time monitoring of critical corporate KPIs with automatic alerts on significant deviations from established baselines and strategic thresholds, complete analysis of customer satisfaction metrics, Net Promoter Score evolution, and loyalty indicators with advanced segmentation to precisely identify loyalty drivers and churn risk factors enabling proactive intervention strategies, or detailed analysis of complete supply chain performance for strategic optimization of inventory levels, supplier relations, and logistical efficiency through identification of substantial improvement areas and significant cost reduction opportunities. This category provides the essential informational base for understanding complex business dynamics and constitutes the indispensable prerequisite for more sophisticated analytics based on this fundamental understanding.

Predictive Analytics Predictive Analytics uses mathematically sophisticated machine learning algorithms, rigorously validated advanced statistical models, and cutting-edge time series analysis techniques to accurately predict future events with measurable confidence intervals, based on complete historical data, patterns identified through rigorous analysis, and statistically significant predictive variables that have demonstrated forecasting power. This advanced discipline requires deep skills in data science, mathematical modeling, sophisticated feature engineering, and rigorous model validation to produce reliable and actionable forecasts that can significantly impact strategic business decisions. Typical but strategically critical and widely validated examples include sophisticated demand forecasting for strategic inventory management optimization, efficient production planning, and fluid supply chain coordination with the ability to incorporate complex external variables like seasonality patterns, emerging market trends, leading economic indicators, and exceptional events that can significantly impact demand patterns, advanced predictive maintenance to anticipate equipment failures, machinery deterioration, and infrastructural problems through sophisticated analysis of real-time sensor data, historical usage patterns, and complete failure data enabling proactive interventions before costly breakdowns occur, advanced credit scoring and sophisticated risk assessment for critical financial evaluations incorporating diverse multiple data sources and predictive behavioral variables to accurately assess creditworthiness and effectively manage portfolio risk, sophisticated churn prediction to identify customers at risk of leaving with sufficient time to enable targeted retention interventions that can significantly impact customer lifetime value and revenue stability, strategic forecasting of market trends and consumer behavior evolution to support strategic product development decisions, marketing investment allocation, and market entry timing that can determine competitive success, sophisticated predictive workforce planning to anticipate hiring needs, training requirements, and optimal resource allocation based on corporate growth projections, seasonal patterns, and skill evolution requirements, or complete financial performance forecasting with detailed scenario analysis and rigorous stress tests for robust strategic planning and proactive risk management supporting informed decision-making in uncertain environments. The accuracy and reliability of these forecasts depend critically on the quality/quantity of underlying data, the mathematical robustness of models employed, and the ability to incorporate relevant domain skills into the modeling process to ensure realistic and business-relevant forecasts.

Prescriptive Analytics Prescriptive Analytics represents the technological apex and strategically most precious of analytical sophistication, combining advanced predictive capabilities with rigorous mathematical optimization algorithms to not only accurately predict what will happen with high probability but also specifically recommend optimal actions to undertake to maximize desired results or minimize risks identified through systematic analysis. This advanced category requires sophisticated integration of mathematically complex disciplines including advanced operations research, rigorous optimization theory, strategic game theory, and applied decision science to create actionable recommendations providing measurable business value. Representative and strategically transformative examples include dynamic optimization of pricing strategies based on real-time demand detection, continuous competitive intelligence, current inventory levels, and customer price sensitivity analysis for systematic revenue maximization while maintaining strong competitive positioning, optimal human resource planning and intelligent scheduling balancing multiple objectives simultaneously including target service levels, operational cost minimization, maintenance of employee satisfaction, and compliance with regulatory constraints through sophisticated mathematical optimization, end-to-end optimization of complex supply chain networks and logistical operations for systematic cost minimization and continuous service optimization coordinating strategic procurement, production planning, distribution networks, and delivery scheduling to achieve measurable operational excellence, automatic and real-time personalization of commercial offers and marketing communications based on customer lifetime value calculations, sophisticated propensity models, and channel optimization algorithms simultaneously maximizing customer value delivery and profitability improvement, advanced recommendation engine systems for intelligent cross-selling and strategic up-selling optimizing customer experience while maximizing revenue opportunities through sophisticated algorithms understanding customer preferences, purchase patterns, and value optimization opportunities, sophisticated optimal asset allocation and mathematical portfolio optimization for financial services with rigorous risk-return optimization and automatic regulatory compliance balancing multiple complex objectives in dynamic market conditions, or complete energy management optimization for manufacturing operations and intelligent building management balancing operating costs, strategic sustainability objectives, and diverse operational requirements through advanced optimization algorithms continuously adapting to changing conditions. Prescriptive Analytics represents the culmination of business intelligence evolution and requires significant investments in specialized talents, advanced technological infrastructure, and sophisticated organizational change management to be implemented effectively and provide measurable value justifying the investment in complexity.

Augmented employee Augmented Employee represents the most sophisticated, strategically innovative, and differentiating category of the Cybernetic Workforce, characterized by the development of highly specialized and personalized AI applications using market-leading Large Language Models accessible via enterprise-grade cloud APIs, advanced RAG (Retrieval Augmented Generation) architectures combining retrieval precision with generation flexibility, high-performance vector databases optimized for semantic search, and the entire modern technological stack necessary to create revolutionary solutions dramatically amplifying cognitive capabilities and operational effectiveness of strategic knowledge workers. This cutting-edge category, while numerically representing a reduced fraction compared to other use cases due to significant development complexity, advanced skill requirements, and sophisticated infrastructure needed, often provides the most significant added value and competitive differentiation for the organization in terms of innovation capacity, sustainable strategic advantage, and unique market positioning that can determine long-term competitive success. Successful development of Augmented Employee solutions requires substantial investments in specialized technical talents, enterprise-grade cloud infrastructure, sophisticated agile development processes, and complete change management to ensure successful adoption and value realization. Representative and transformative examples include the development of highly specialized AI assistants for specific vertical domains such as complete legal AI for automatic contract analysis, due diligence acceleration, and regulatory compliance automation understanding complex legal terminology, case law precedents, and jurisdiction-specific requirements to effectively support law firms, corporate legal departments, and professional compliance organizations, sophisticated medical AI for advanced diagnostic support, evidence-based treatment recommendations, and intelligent clinical decision support incorporating vast medical knowledge, complex drug interactions, and current evidence-based medicine principles to support professional healthcare providers and medical researchers at the forefront, advanced financial AI for sophisticated risk assessment, complete investment analysis, and intelligent algorithmic trading leveraging real-time market data, leading economic indicators, and proprietary mathematical models to support competitive financial institutions, strategic investment firms, and effective regulatory bodies, creation of revolutionary knowledge management systems combining advanced RAG technologies with accumulated organizational skills for institutional knowledge preservation, significant onboarding acceleration, and distributed decision support enabling knowledge democratization in the organization while maintaining high accuracy and consistency, development of sophisticated content generation platforms for excellence in marketing, professional communications, and complete technical documentation maintaining consistent brand voice, automatic regulatory compliance, and enterprise quality standards while enabling significant creativity improvement and productivity acceleration, realization of advanced decision support systems integrating multiple internal and external data sources with proprietary AI models for sophisticated strategic planning, actionable competitive intelligence, and complete scenario analysis supporting informed decision-making in complex environments, creation of revolutionary training environments and advanced simulation platforms using AI for personalized learning paths, adaptive intelligent assessment, and precise skill gap analysis to accelerate professional development and systematically improve organizational capabilities, development of sophisticated customer service AI combining deep product knowledge, complete customer history, and artificial emotional intelligence to provide superior customer experience building loyalty and significantly increasing satisfaction metrics, or realization of intelligent research assistants and development support tools accelerating innovation cycles through automatic literature analysis, creative hypothesis generation, and mathematical experimental design optimization that can significantly reduce time-to-market and improve innovation success rates. Augmented Employee requires strategic investments in advanced technical skills, modern cloud infrastructure, sophisticated agile development methodologies, and complete change management programs, but offers the most radical transformation potential for knowledge-intensive processes and the possibility to create sustainable competitive advantages difficult for competitors to replicate effectively.

4.4 Domain "Tools"

The tools dimension represents the fundamental technological infrastructure enabling the concrete operational realization of identified use cases, articulating itself according to a strategic logic reflecting different implementation approaches with significant implications in terms of investments, required skills, realization times, and necessary governance. The Make, Build-Configure, and Buy classification is not simply a technical categorization but reflects different philosophies of technological innovation management, vendor management strategies, risk management approaches, and total cost of ownership models that the organization must carefully consider in building its AI ecosystem. The variety of tools available in each category suggests the measure of intrinsic complexity of the ecosystem, where every specific nature of use case can be articulated on one or sometimes more different tools simultaneously, requiring specific skills, differentiated management approaches, and sophisticated integration strategies to maximize synergies and minimize operational and strategic conflicts.

4.4.1 Make - Internal development

The Make category represents the most ambitious and strategically autonomous approach, characterized by maximum customization flexibility and complete control over the technological roadmap, but also the most complex from the point of view of required technical skills, necessary governance, and investments in talent and infrastructure. This mode implies complete internal development of solutions using fundamental technological components that are orchestrated, integrated, and customized to create specific applications perfectly aligned with the unique needs of the organization and its distinctive processes. The Make approach requires advanced software engineering capabilities, AI and machine learning engineering skills, sophisticated infrastructure management, and dedicated change management, but offers the potential to create truly differentiating solutions and sustainable competitive advantages that can determine long-term competitive success.

Applications The development of customized applications represents the most ambitious and complete mode of AI implementation, requiring complete and integrated skills in software development, AI and machine learning engineering, user experience design, and domain skills to create end-to-end solutions solving specific business problems with innovative and differentiating approaches. Custom application development requires highly qualified multidisciplinary teams, sophisticated agile development methodologies, enterprise-level development and operations infrastructure, and significant investments in quality assurance and user acceptance testing to ensure robust, scalable solutions perfectly aligned with strategic business requirements. This approach allows the creation of completely bespoke systems that integrate perfectly into existing operational processes, support organization-specific workflows, and incorporate proprietary business logics reflecting the company's unique skills and competitive differentiators.

Algorithms The algorithms category comprises the internal development of proprietary machine learning models, advanced mathematical optimization algorithms, and specific processing logics representing the organization's main technological intellectual property and often constituting the basis for sustainable competitive advantages in the reference market. The development of proprietary algorithms requires advanced skills in mathematical modeling, rigorous statistical analysis, machine learning theory, and deep domain competence to create algorithms exceeding the performance of standard commercial solutions and perfectly adapting to the organization's specific data, processes, and objectives. This category includes the development of customized classification algorithms for automatic categorization of corporate content, proprietary predictive models based on organization-specific historical data, matching and similarity algorithms for corporate search systems, optimization algorithms for supply chain management and resource allocation, customized natural language processing algorithms for sector-specific terminology, or computer vision algorithms for quality control and process monitoring specifically trained on the organization's products and processes.

GUI (Graphical User Interface) The development of customized graphical interfaces represents a strategic investment to create optimized user experiences maximizing adoption, productivity, and user satisfaction through design centered on the specificities of corporate workflows, end-user skills, and specific corporate objectives. In this context fall the creation of customized executive dashboards presenting key performance indicators and insights in formats specific to management, development of simplified interfaces for non-technical operators who need to interact with complex AI systems, realization of mobile applications for field workers and remote employees, creation of interactive training and onboarding interfaces, and significantly also tools for creating customized chatbots, conversational design platforms, and development frameworks for virtual assistants allowing the realization of conversational interfaces tailored to specific organizational needs, incorporating corporate terminology, internal processes, and proprietary business logics to create conversational experiences reflecting the organization's identity and values. The importance of this component goes beyond "display preferences" of content or interaction modes: it is also important to guarantee transparency and be able to properly route output to underlying systems.

API (Application Programming Interface) In the Make context, APIs mainly represent programmatic access and integration with Large Language Models and external AI services through standardized interfaces like OpenAI API, Anthropic Claude API, Google AI API, or Azure OpenAI Service allowing direct integration of advanced generative capabilities into customized applications developed internally. This includes implementation of API calls for text generation, completion, embedding, moderation, and fine-tuning, management of authentication, call rate limiting and error handling, implementation of caching strategies to optimize performance and costs, creation of custom wrappers and middleware to standardize access to different providers, development of unified interfaces abstracting differences between different models and providers to guarantee flexibility and resilience in application architecture, and integration of these services into complex corporate workflows requiring sophisticated orchestration and state management to support advanced use cases going beyond simple request-response calls.

RAG (Retrieval Augmented Generation) The implementation of personalized and sophisticated RAG architectures represents one of the most strategic and high-value approaches to valorize corporate information assets accumulated over time through modern generative technologies, creating knowledge systems combining the vastness and accuracy of proprietary databases with the flexibility and intelligence of Large Language Models for business-critical use cases. This includes realization of advanced RAG systems for corporate knowledge management combining technical documentation, organizational best practices, and corporate skills with natural language query capabilities, development of specialized AI assistants accessing proprietary databases and customer information in real-time to provide contextual support, creation of Q&A systems for customer support integrating product documentation and support ticket history, implementation of research assistants combining scientific literature with internal experimental data, realization of compliance and regulatory guidance systems integrating legal documents and internal policies, or development of training platforms combining learning materials with corporate case studies. Successful implementation of RAG systems requires skills in vector databases, embedding technologies, prompt engineering, and domain-specific optimization to ensure accuracy, relevance, and optimal performance.

ML (Machine Learning) Internal development of complete and end-to-end machine learning pipelines represents the most technically intensive investment but also the one with the greatest potential to create intellectual property constituting sustainable competitive advantage through the creation of models specifically trained on the organization's data, processes, and objectives. This category comprises the entire machine learning lifecycle from data preparation and feature engineering to model training, validation, distribution, and monitoring in production environments. This includes creation of proprietary computer vision systems for quality control in manufacturing specifically trained on organization's products, development of customized natural language processing models for customer feedback analysis and brand monitoring, realization of anomaly detection systems for cybersecurity and operational monitoring learning normal behavior patterns specific to the organization, creation of ensemble models combining multiple machine learning techniques for business-critical predictions, development of reinforcement learning systems for optimization problems like dynamic pricing and resource allocation, or implementation of deep learning architectures for complex pattern recognition specifically designed for the organization's data types. Successful development of machine learning requires data science skills, machine learning operations capabilities, scalable computational infrastructure, and governance frameworks to ensure reliability, fairness, and explainability of models.

Analytics Development of customized and complete analytics platforms represents the most strategic investment to create data-based decision-making capabilities transforming corporate data assets into competitive intelligence, operational insights, and strategic guidance through infrastructure for data processing, statistical analysis, and visualization optimized for the organization's specific information needs. This includes realization of enterprise-level data lakes with real-time processing capabilities for advanced business intelligence, realization of analytics platforms for strategic planning and risk management. These platforms require skills in big data technologies, statistical modeling, data visualization, and business intelligence to provide actionable insights generating measurable business value.

4.4.2 Build-Configure - Platform customization

The Build-Configure category represents a balanced strategic approach combining customization flexibility with the stability, reliability, and support ecosystem of consolidated and mature technological platforms. This mode allows organizations to obtain sophisticated and customized solutions without having to develop everything from scratch, significantly reducing time to market, technical risk, and total cost of ownership while maintaining sufficient flexibility to address specific business requirements and differentiating use cases. The Build-Configure approach requires specific configuration, integration, and customization skills but does not necessitate the entire stack of capabilities required for complete internal development, allowing organizations to concentrate resources on key differentiators while leveraging R&D investments of specialized vendors.

Generalists: Chatbot In the Build-Configure context, generalist chatbots refer to direct access and extensive use of consolidated and mature conversational platforms like ChatGPT, Claude, Gemini, Copilot, and other generative AI solutions accessible through standard web interfaces, dedicated applications, or corporate integrations providing immediate conversational capabilities without requiring complex development or advanced technical configuration. These tools represent the most immediate and accessible mode to exploit advanced generative AI capabilities, allowing users to interact directly with state-of-the-art language models for a vast range of cognitive tasks through prompt engineering, guided conversation techniques, and interaction design methodologies maximizing effectiveness of generated responses. Use of generalist chatbots requires skills in prompt engineering, conversation management, and understanding of specific capabilities and limitations of each platform to optimize results and ensure appropriate usage according to corporate policies and compliance requirements, but offers immediate access to sophisticated capabilities without investments in infrastructure or internal development.

Generalists: Agent Agents in the generalist context represent agentic and semi-autonomous use of consolidated AI tools that can operate with greater temporal autonomy, memory persistence, and multi-phase execution capability compared to simple synchronous conversational interactions. Representative examples include platforms like Copilot Studio for creating conversational agents based on proprietary data domains, Manus for research automation and content curation, Genspark for intelligent search and information synthesis, Perplexity Pro for advanced search capabilities with source citation, OpenAI's custom GPTs allowing specific configurations and personalized knowledge bases, Claude's configurable agents maintaining context across multiple sessions, or workflow-enabled agents integrating with productivity and collaboration platforms to automate complex tasks requiring interaction with multiple data sources and contextual decision-making. These agents represent a significant evolution compared to traditional chatbots as they can maintain state, access external tools, execute asynchronous operations, and provide results persisting beyond single interaction session, enabling more sophisticated use cases requiring continuity and integration with existing corporate workflows.

Generalists: Cloud Svcs Generalist Cloud Services in the Build-Configure context specifically refer to programmatic and scalable access to linguistic platforms through standardized APIs and managed cloud services like OpenAI API for GPT model access, Anthropic Claude API for advanced conversational capabilities, Google AI API for integration with Gemini and Google AI services, Azure OpenAI Service for enterprise distribution of OpenAI models providing reliable and scalable access to advanced language models with fluid integration capability into corporate applications, workflow systems, and automation platforms through standard API calls supporting authentication, rate management, monitoring, and granular billing. These services offer the advantage of automatic scalability, high availability, continuous model updates, and enterprise-level support without requiring investments in specialized computational infrastructure or skills in model distribution and management, allowing organizations to focus on integration logic and business value rather than infrastructure management and model operations.

Platform: RPA Corporate RPA platforms represent mature and complete solutions for process automation combining intuitive visual development environments, robust and scalable execution engines, and centralized management capabilities through consolidated and battle-tested tools like UiPath Studio for advanced automation workflow development, Blue Prism for regulated industries and high-security environments, or WorkFusion for intelligent automation combining RPA with AI capabilities. These platforms provide drag-and-drop design interfaces to democratize automation development, extensive pre-built connectors for integration with legacy systems and modern applications, exception management capabilities for robust error handling, scalable execution infrastructure for high-volume processing, comprehensive monitoring and analytics for performance optimization, and centralized governance for security and compliance management. Corporate RPA platforms allow developers to create, test, distribute, and manage automation workflows interacting with diverse systems, manipulating structured and unstructured data, executing complex business logic, and integrating fluidly with existing corporate architecture, offering rapid distribution, measurable ROI, and significant risk mitigation through process standardization and error reduction.

Platform: ADA (Advanced Desktop Automation) ADA represents a specialized and strategically important category of automation tools like Microsoft Power Automate Desktop for Windows automation, UiPath StudioX for simplified desktop automation, or customized desktop automation frameworks specifically designed and optimized to automate knowledge-intensive "desktop" work processes involving complex interaction with application interfaces, local file systems, email clients, productivity tools, and software applications used by knowledge workers in their daily work routines. These powerful tools allow creating sophisticated automation workflows operating directly on desktop environments, accurately simulating human interactions with software applications through screen scraping technologies, advanced UI element recognition, precise keyboard and mouse automation, intelligent wait strategies managing application response variability, and robust error handling to manage unforeseen conditions. ADA tools are particularly precious for organizations where significant knowledge work still relies on desktop applications not offering robust API interfaces or where integration with legacy systems through user interfaces represents the most feasible and economical automation approach, allowing automation of data entry workflows, report generation processes, document processing operations, and communication management tasks requiring interaction with multiple desktop applications simultaneously.

4.4.3 Buy - Solution acquisition

The Buy category represents a pragmatic and time-efficient approach to access advanced AI capabilities through the acquisition of complete, mature, and widely tested solutions minimizing time to market, reducing implementation risk, and leveraging consolidated skills and continuous innovation investments of specialized vendors dedicating significant resources to R&D and innovation. This mode offers less customization flexibility but provides immediate access to proven technologies, extensive support ecosystems, regular updates keeping solutions up to date with latest technological advancements and security standards, and professional services accelerating distribution and adoption. The Buy approach is particularly strategic when adoption speed might result critical towards competitive advantage, when internal development skills are limited or expensive to acquire, or when reliable vendor solutions provide sufficient capacity to address business requirements without justifying significant internal development investments that could distract resources from main business activities.

AI Extensions: from Vendors AI solutions provided directly by existing corporate system vendors represent the smoothest and strategically most solid path to enhance consolidated corporate processes with AI capabilities since these modules are natively designed and carefully tested to integrate perfectly with existing system architectures, consolidated data models, and operational business workflows maintaining all security characteristics, compliance standards, and support ecosystem of the main system. The main advantage of vendor solutions is immediate availability without extended procurement cycles, guaranteed compatibility eliminating integration risks, simplified governance inheriting existing policies and controls, reduced integration complexity minimizing technical debt, continuous support ensuring continuity and reliability, and regular updates keeping capabilities aligned with product roadmaps and sector standards without requiring internal resources for maintenance and evolution. The limits lie in the natural positioning "at the extremes" of the customization range of these solutions: perfectly generalist or radically specific.

AI Extensions: from System Integrators Specialized solutions implemented by system integrators represent a strategically precious hybrid approach combining proven technological components with specialized implementation skills, sector-specific customizations, and deep domain knowledge accumulated through multiple successful implementations in similar contexts and especially in the specific corporate context (if speaking of system integrators used to working with the company, perhaps as implementers or maintainers of foundational systems). System integrator approaches provide access to specialized skills that can be expensive to develop internally, proven methodologies reducing implementation risks, continuous support ensuring long-term success, and customization capabilities addressing unique business requirements that generic vendor solutions cannot adequately support.

Verticals: Fine-tuned LLM Large Language Models fine-tuned for specific sectors represent an emerging and strategically important category of AI solutions leveraging fundamental capabilities of general-purpose language models while adding specialized knowledge, sector-specific terminology, and reasoning capabilities adapted for industries, professions, or domain applications through additional training on domain-specific datasets, use case optimization, and performance tuning resulting in superior accuracy and contextual appropriateness for professional use cases. These specialized models combine broad linguistic understanding and reasoning capabilities of foundation models with deep skills in specialized fields to provide more accurate, relevant, and contextually appropriate answers for professional workflows requiring domain skills, regulatory compliance, and sector-specific best practices, offering significant value for organizations operating in regulated industries or high-competence domains where generic models might not satisfy accuracy, compliance, or performance requirements necessary for mission-critical applications.

Verticals: Stand Alone AI App Stand-alone AI applications represent highly specialized tools concentrating on specific use cases or functional domains with deep skills, advanced optimization, and superior performance in their designated domains, following the philosophy of "doing one thing exceptionally well" rather than providing broad AI capabilities that might result less effective for specific tasks. Representative and widely adopted examples include GAMMA for automated presentation creation generating professional quality presentations through AI-enhanced content design and optimization, MIDJOURNEY for AI-enhanced image generation producing artistic and commercial quality visual content, Jasper for marketing content creation specializing in brand-coherent copy generation, Grammarly for advanced writing assistance combining grammar correction with style improvement and tone optimization, Otter.ai for meeting transcription and intelligent summary automating documentation and follow-up, Synthesia for AI video creation with realistic avatars, Copy.ai for complete marketing content generation, Whisper for accurate voice-to-text transcription, Figma AI for design automation and creative assistance, Tableau AI for automated data analysis and insight generation. These tools are particularly precious for organizations wishing immediate access to cutting-edge AI capabilities for specific functions without significant implementation overhead, technical complexity, or extensive training requirements, offering rapid distribution, proven effectiveness, and specialized performance often exceeding general-purpose solutions for their designated use cases.

Verticals: Domain SLM Domain Small Language Models represent a strategically specialized approach utilizing smaller and more focused AI models that are specifically trained or extensively fine-tuned on highly specific datasets for particular industries, professional domains, or organizational contexts to provide expert-level performance for narrow use cases while requiring significantly less computational resources and offering greater control over model behavior, accuracy standards, and regulatory compliance compared to large general-purpose models that might be excessive for specific applications. Targeted and professional quality examples include legal SLMs developed specifically for patent analysis, contract review, and legal precedent research providing specialized capabilities for intellectual property law, corporate law, and litigation support, accounting and finance SLMs optimized for tax preparation, financial statement analysis, and audit procedures incorporating accounting standards and regulatory requirements, medical SLMs specialized for clinical documentation, diagnostic coding, and treatment protocol recommendations understanding medical terminology and clinical guidelines, engineering SLMs focused on technical specification analysis and design optimization for specific industries, or customer service SLMs trained on specific company products and procedures maintaining brand voice and ensuring compliance. The key advantage of SLMs is focused skills providing superior accuracy for specific domains, reduced computational requirements reducing operating costs, greater interpretability supporting regulatory compliance, easier customization enabling organizational adaptation, and simplified compliance management reducing governance complexity, while potential limitations include narrower capability scope and periodic retraining requirements as domain knowledge evolves.

Embedded AI The Embedded AI category represents the native and fluid integration of AI functionalities directly into productivity tools and work platforms that employees already use daily, representing perhaps the most impactful and strategically significant approach for widespread AI adoption within organizations as it minimizes training requirements, significantly reduces change management complexity, and dramatically accelerates adoption rates through familiar interfaces and integration of existing workflows. This strategic approach eliminates adoption barriers often characterizing new technology implementations, allowing users to access AI functionalities within familiar interfaces, consolidated workflows, and trusted platforms without requiring extensive learning curves or workflow interruptions that could impact productivity during transition periods. Key examples and market leaders include Microsoft Copilot seamlessly integrated throughout the Office 365 suite to provide writing assistance in Word, data analysis capabilities in Excel, presentation creation automation in PowerPoint, and meeting summaries in Teams creating a complete AI-enhanced productivity environment, Google Gemini strategically incorporated into Google Workspace for intelligent email composition in Gmail, document analysis improvement in Google Docs, data insight generation in Google Sheets, and meeting intelligence in Google Meet, Adobe AI capabilities integrated throughout Creative Suite for automation of content-aware editing, intelligent crop optimization, automated color correction, and style transfer capabilities improving creative workflows without requiring artists to learn complex separate AI tools. Other significant examples include Slack AI for intelligent conversation summary, improved search capabilities, and automated workflow triggers improving team communication efficiency, Zoom AI for meeting transcription accuracy, automated action item extraction, and participant engagement analysis substantially improving meeting productivity, Notion AI for knowledge management improvement and content creation assistance, Superhuman AI for email management intelligence, Loom AI for video communication optimization, or Canva AI for design automation democratization. In this strategically important category also fall sophisticated tools for creating personalized agents fluidly integrated into existing platforms, like Copilot Studio for Microsoft 365 ecosystem enabling creation of custom agents integrating with Office applications, Power Platform automation tools allowing development of agents with extensive integration capabilities, or configurable agents within collaboration platforms maintaining context and providing intelligent assistance within existing workflow frameworks users already understand and trust.

Agent The final category under the Buy approach constitutes specialized and pre-built agents that can be acquired and configured for specific organizational functions rather than developed internally, representing intelligent automation solutions combining conversational AI capabilities, sophisticated workflow orchestration, and consolidated domain skills to address common business use cases across diverse industries with proven effectiveness and rapid distribution capabilities. These agents represent the culmination of vendor skills in specific functional areas, offering immediate access to sophisticated capabilities that would require extensive development time, specialized skills, and significant investments if developed internally, providing at the same time continuous vendor support, regular updates, and proven functionality substantially reducing implementation risk. These purchased agents offer rapid distribution accelerating time to value, proven functionality reducing technical risk, continuous vendor support ensuring ongoing effectiveness, regular updates maintaining currency with technological advancements, and configuration flexibility allowing adaptation to specific organizational requirements, necessary system integrations, and unique business processes, while considerations include vendor dependence for ongoing support, customization limitations potentially restricting perfect fit, ongoing subscription costs impacting total cost of ownership, and potential integration challenges with existing systems possibly requiring additional development effort.

5 Cross-domain activation map

The aforementioned complexity and articulation of the ecosystem, which sees the single use case implementable with different technological solutions, based on different data morphologies and activating complex relationships with the layer of foundational systems, requires detailed analysis (however slightly repetitive) of the different typologies of elements activated by each class. In the paragraphs that follow I will carry out this analysis unrolling for each use case typology all components (data, tools, resources, and systems) that are (or can be) "activated" by them.

5.1 Personal assistant - synchronous

PERSONAL ASSISTANT - SYNCHRONOUS type Use Cases are typically realized using generalist chatbots or agentic functions of the same, or using specialized vertical systems (Stand Alone Apps). They access and interact with Collaboration and personal productivity systems (Mail, messaging systems, File Systems etc.) through which they utilize structured, semi-structured, and unstructured data producing structured and unstructured data as output. To maximize their impact they must be able to access definitional taxonomies (hierarchies and structures of data, documents) and their use must be governed by policies. Use cases thus implemented must be documented to be usable and since most of these use cases are developed "by prompt" the use of the Prompt Library resource organizing and structuring efficient and tested interrogation structures is important. The development of efficient prompts allowing democratically the entire company to autonomously approach this type of use case must be part of specific training and materials must be accessible and available via the AI Portal.

5.2 Personal assistant - asynchronous

PERSONAL ASSISTANT – ASYNCHRONOUS type Use Cases are typically realized through specialized personal automation tools (ADA - Personal Automation) or via dedicated vertical tools (AI App Stand Alone). These tools operate asynchronously and have a significantly broad interaction perimeter: besides typical personal productivity and collaboration systems (Mail, messaging, shared file systems), they access, via ADA interface, directly also more structured and complex foundational systems like ERP, CRM, SRM, transactional systems and MIS, Databases and Data Warehouses (DWH), Business Intelligence (BI), Corporate Performance Management (CPM), MES, CAD/CAM, E-commerce/Biz systems and "field" systems. This extended integration allows extraction, manipulation, and generation of prevalently structured data, even if not exclusively, through automated asynchronous workflows. Optimal management of such data requires direct and structured access to definitional taxonomies (hierarchies and classifications of data/documents) and is disciplined by specific corporate policies to guarantee governance and compliance. Implemented Use Cases must be adequately documented and cataloged to guarantee usability and replicability at the corporate level. Given the central role that prompt definition covers in ADA workflows, the use of a Prompt Library maintaining validated, efficient, and consistent interrogation structures is fundamental. From an organizational and training point of view, widespread enablement of corporate users requires specific training materials, made easily available via the AI Portal.

5.3 Enhanced use

ENHANCED USE type Use Cases are characterized by the prevalent use of AI Extensions coming from specialized vendors or realized through the intervention of system integrators, as well as Embedded AI systems directly integrated into existing corporate applications. This integration typology generally occurs via vertical fine-tuned Large Language Model (LLM) models and dedicated AI tools, inserted within application platforms adopted by the organization. Enhanced Uses interface with a wide range of foundational systems such as ERP, CRM, transactional systems and MIS, Databases and Data Warehouses, BI, CPM, MES, CAD/CAM, E-commerce and field systems. Through these systems, use cases access structured, semi-structured, and unstructured data present in corporate applications, further transforming them into structured and unstructured data suitable for use by AI. To guarantee effective implementation of Enhanced Use Use Cases, it is indispensable to have precise definitional taxonomies (hierarchies, classifications, and data structures) and a solid reference to consolidated best practices. It is equally necessary to adopt specific policies defining clearly operational and organizational methods, supporting users with complete and structured documentation of use cases (Use Case Doc).

5.4 Automation - RPA

AUTOMATION - RPA type Use Cases are typically realized through specialized RPA (Robotic Process Automation) platforms operating in automated mode. These tools interact with an extended perimeter of foundational systems including ERP, CRM, SRM, Database/DWH, BI, CPM, MES, CAD/CAM, E-COM/BIZ, and Collaboration & personal productivity systems such as Mail, Messaging, Shared File Systems, Calendars, Personal Prod. From a data point of view, RPA operates prevalently on structured and semi-structured data, transforming them into structured and unstructured data through automated processes. To guarantee effective implementation, RPA use cases require access to structured definitional taxonomies and must be governed by specific corporate policies. Their corporate usability depends on the availability of appropriate training materials and systematic documentation through Use Case Doc guaranteeing replicability and governance.

5.5 Automation - Intelligent automation

AUTOMATION - INTELLIGENT AUTOMATION type Use Cases are typically realized through RPA platforms enhanced by AI components and CLOUD SVCS tools for intelligent process processing. These tools interact with an extended perimeter of foundational systems including ERP, CRM, SRM, Database/DWH, BI, CPM, MES, CAD/CAM, E-COM/BIZ, and Collaboration & personal productivity systems such as Mail, Messaging, Shared File Systems, Calendars, Personal Prod. From a data point of view, Intelligent Automation operates on structured and semi-structured data, transforming them into structured and unstructured data through intelligent automated processes. To guarantee effective and scalable implementation, Intelligent Automation use cases require systematic access to well-structured definitional taxonomies and must be rigorously governed by specific corporate policies. Their usability and replicability at corporate level depend on the availability of specialized training materials and systematic documentation through Use Case Doc guaranteeing governance and operational compliance.

5.6 Analytics

ANALYTICS type Use Cases are typically realized through Machine Learning (ML) algorithms and specialized ANALYTICS platforms for corporate data processing and analysis. These tools interact with an extended perimeter of foundational systems including ERP, CRM, SRM, Database/DWH, BI, CPM, MES, CAD/CAM, E-COM/BIZ, and Collaboration & personal productivity systems such as Mail, Messaging, Shared File Systems, Calendars, Personal Prod. From a data point of view, Analytics operate exclusively on structured data, transforming them through advanced analytical processes into new structured data. For their optimal functioning, they require access to well-structured definitional taxonomies organizing and classifying input data, as well as a SOURCE CODE library containing consolidated algorithms and analytical models. To guarantee effective and accurate implementation, Analytics use cases must be rigorously governed by specific corporate policies defining methodologies and quality standards. Their usability and scalability at corporate level depend on the availability of specialized training materials and systematic documentation through Use Case Doc guaranteeing governance, reproducibility, and operational compliance.

5.7 Augmented employee

AUGMENTED EMPLOYEE type Use Cases are typically realized through applications developed with GUI interfaces, APIs, and RAG (Retrieval-Augmented Generation), supported by CLOUD SVCS services and specialized Fine-tuned LLM and domain SLM solutions, as well as Stand Alone AI Apps and dedicated AGENT systems. These tools interact with an extended perimeter of foundational systems including ERP, CRM, SRM, Database/DWH, BI, CPM, MES, CAD/CAM, E-COM/BIZ, and Collaboration & personal productivity systems such as Mail, Messaging, Shared File Systems, Calendars, Personal Prod. From a data point of view, Augmented Employee operates on structured, semi-structured, and unstructured data, applying tagging processes for their classification and transforming them through embedding techniques into structured, unstructured, and vectorized data. To guarantee effective and scalable implementation, Augmented Employee use cases require systematic access to definitional taxonomies, optimized prompt libraries, and consolidated source code libraries. They must be rigorously governed by specific corporate policies defining operational methods and security standards. Their usability and widespread adoption at corporate level depend on the availability of specialized training materials and systematic documentation through Use Case Doc guaranteeing governance, reproducibility, and operational compliance.

6 The Corporate AI Portal: logical and functional architecture of democratic access to the ecosystem

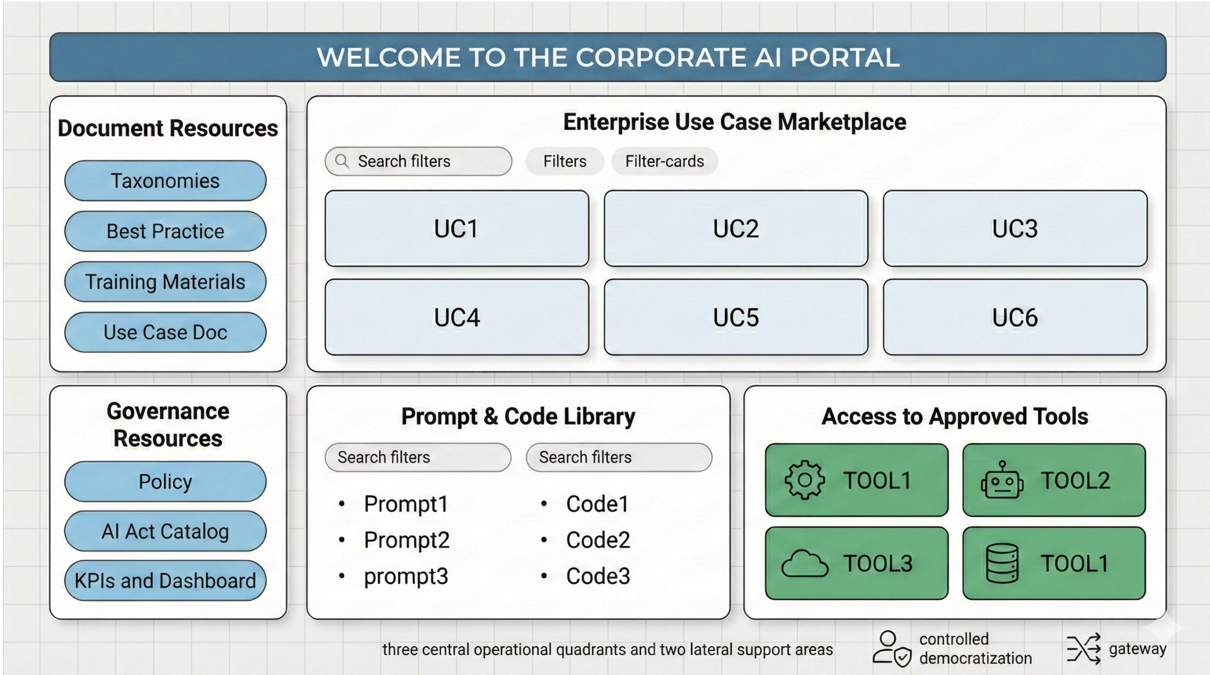

Figure 3: Logical Architecture of the Corporate AI Portal

The effective implementation of a corporate AI ecosystem requires not only the definition of sophisticated conceptual frameworks and the orchestration of complex technological components, but also the realization of access interfaces democratizing the use of AI resources throughout the entire organization. The AI Portal represents the materialization of this strategic need: a unified digital environment acting as an intelligent gateway between human skills and artificial capabilities, transforming the intrinsic complexity of the ecosystem into an accessible, governed, and strategically aligned user experience. Empirical research on the most mature AI implementations highlights that long-term success of AI adoption depends critically on the organization's ability to overcome the paradox of "controlled democratization": making AI technologies accessible to the largest possible number of internal users while simultaneously maintaining rigorous standards of governance, security, and result quality. The AI Portal responds to this challenge through an architecture combining access flexibility with systematic control, user experience personalization with process standardization, and operational autonomy with strategic alignment.

6.1 Logical architecture: the five domains of AI experience

The Portal architecture is articulated according to a logic comprising three central operational quadrants and two lateral support areas, reflecting different modes through which users interact with the AI ecosystem, each characterized by specific functional needs, levels of technical complexity, and degrees of operational autonomy. The structure is completed with the two lateral areas dedicated to resources transversally supporting the entire ecosystem: Documentary Resources and Governance Resources.

6.1.1 Use case marketplace: the catalog of applied intelligence

The first quadrant constitutes the strategic heart of the portal, representing the materialization of application knowledge assets accumulated by the organization in its AI adoption journey. The Use Case Marketplace is configured as a dynamic and intelligently organized repository presenting all use cases implemented, tested, and validated by the organization through an interface privileging guided discovery and functional accessibility over underlying technical complexity. The marketplace structure should reflect the taxonomy of use cases defined in the framework: Personal Assistant (Synchronous and Asynchronous), Enhanced Use, Automation (RPA and Intelligent Automation), Analytics (Descriptive, Predictive, Prescriptive) and Augmented Employee. Each category is presented through informative cards providing a synthetic but complete description of the use case (also retrievable in documentary access to use case documentation, see later), indications on expected benefits, usage requirements, and direct links to necessary tools for access and use (e.g., internal cost allocation policies, system requirements, people to contact within the organization as "case owners" etc). The intelligence of the cataloging system lies in its ability to offer sophisticated search and filtering mechanisms allowing users to navigate the use case heritage according to different dimensions: type of corporate process involved, implementation complexity level, expected organizational impact, skills required for use, and alignment with specific strategic objectives. This multi-dimensional structuring transforms the search for AI solutions from a potentially dispersive exploratory process into a guided discovery path of the most relevant opportunities for the user's specific context.

6.1.2 Prompt & code library: the arsenal of intelligent interaction

The second quadrant represents the practical arsenal enabling effective interaction with AI technologies, systematically organizing all assets of procedural and technical nature supporting operational implementation of use cases. This section is articulated in two complementary components reflecting the two main modes of interaction with AI systems: the conversational one based on prompts and the programmatic one based on code.

The Prompt Library constitutes a strategic resource collecting, organizing, and making searchable all interrogation and instruction models optimized for different AI tools used by the organization. This library is not simply a collection of texts, but an intelligent procedural knowledge management system classifying prompts according to multiple dimensions: application use case, destination AI tool, complexity level, typical results obtained, and variants for specific contexts. Systematic organization of prompts significantly accelerates users' learning curve, guarantees consistency in results obtained across different teams and work sessions, and facilitates sharing of best practices for AI interaction. The creation of the library becomes a shared and widespread exercise of contribution and training and should constitute the first operational piece of developing a "corporate" resource.

The Code Library organizes all source code assets developed internally for customizations, integrations, and proprietary solutions, constituting the basis for component reuse, acceleration of new solution development, and technical skill sharing across the organization. This library includes not only source code but also associated technical documentation, usage examples, validation tests, and indications for evolution and maintenance of algorithms developed for different purposes. Both components are organized according to advanced search systems allowing users to rapidly identify the most relevant resources for their specific context, reducing time needed for implementing new use cases and improving quality of results obtained through reuse of validated solutions.

6.1.3 Access to approved tools: the controlled technological gateway

The third quadrant materializes the direct access interface to AI tools approved by the organization, transforming the complexity of technological selection and configuration into a simplified user experience simultaneously maintaining all necessary governance and security controls. This section represents the practical translation of strategic decisions taken by the AI Center of Excellence regarding tool selection, their corporate configuration, and access modes for different user typologies. In these tools are "tendentially" represented mainly generalist tools, verticals, and/or agents developed on them (leaving to the Use Case marketplace the role of "hiding" the technical complexity of other tool typologies). Each tool is presented through informative cards including functional description, typical use cases, usage requirements, links for direct access, support documentation, and information on available internal support mechanisms. The architecture of this quadrant should incorporate Single Sign-On (SSO) mechanisms allowing authorized users to access tools directly without having to manage multiple credentials, role-based authorization systems guaranteeing that each user has access only to appropriate tools for their responsibilities and skills, and audit mechanisms systematically recording tool usage for governance and optimization purposes.

6.1.4 Lateral support areas: ecosystem foundations

The portal architecture is completed with two lateral areas providing the informational and normative substrate necessary for effective functioning of the entire corporate AI ecosystem.

Documentary resources: ecosystem cognitive infrastructure The Documentary Resources area represents the informational backbone supporting the entire AI ecosystem, organizing and making accessible all metadata resources constituting the knowledge base necessary for effective and compliant use of AI technologies. This section articulates through four main categories reflecting different typologies of knowledge necessary to support AI operations at all organizational levels. **Taxonomies** constitute the classificatory nervous system of the ecosystem, providing standardized conceptual structures for organizing data, documents, processes, and skills. This resource is particularly critical to ensure semantic consistency across different use cases and tools, enabling interoperability and facilitating systematic governance of the entire corporate information asset. **Best Practices** represent the distillation of organizational experience, collecting consolidated methodologies, verified approaches, and solutions to recurring problems emerging in implementation and daily use of AI technologies. This category includes both technical best practices (optimal configurations, effective parameterizations, integration modes) and organizational ones (change management processes, user engagement strategies, feedback and continuous improvement mechanisms). Of particular strategic relevance are usage instructions provided by vendors and system integrators developing AI solutions integrated into foundational corporate systems: these specialized guides constitute a critical asset facilitating optimal adoption of AI functionalities native to ERP, CRM, BI, and other corporate platforms, ensuring the organization can maximize value of investments in proprietary AI extensions from providers. **Training Materials** systematically organize all educational content necessary to develop and maintain AI skills at all organizational levels, from basic training on understanding AI technologies to advanced specialization on specific tools and methodologies. This category is designed to support differentiated training paths responding to diverse skill needs present in the organization. **Use Case Documentation** constitutes the technical repository systematizing all information necessary to understand, replicate, and modify implemented use cases, including technical specifications, configurations, obtained results, lessons learned, and recommendations for future evolution.

Governance resources: normative and control framework The Governance Resources area materializes the system of controls, standards, and supervision mechanisms ensuring responsible, compliant, and strategically aligned use of AI technologies. This section represents the operational translation of AI Strategic Committee decisions and provides users with all tools necessary to operate within parameters defined by the organization. **Policies** define the internal normative framework governing AI use, establishing ethical principles, operational limitations, individual and organizational responsibilities, and compliance procedures. This dimension is particularly critical in the current regulatory context characterized by the European AI Act and growing sensitivity towards responsible use of artificial intelligence in the corporate sphere. Policies are not limited to defining prohibitions and limitations, but provide positive guidance for optimal use of AI technologies in compliance with organizational values and strategic objectives. The **AI Act Catalog** represents a specialized resource systematically organizing all information necessary to ensure compliance with European artificial intelligence regulation. This section includes risk assessment tools (models, questionnaires etc), compliance procedures, templates for required documentation, and guidance for AI system classification according to regulatory categories. The catalog serves both as an operational tool for teams implementing AI solutions and as an informational resource for management who must make strategic decisions in a complex and evolving regulatory context. **KPIs and Dashboards** constitute the measurement and monitoring system allowing AI governance to maintain visibility on effectiveness, impact, and compliance of AI implementations across the organization. This section includes tool usage metrics, use case performance indicators, ROI and business impact measures, and executive dashboards supporting strategic decision-making. The KPI system is designed to provide both detailed operational visibility for technical teams and strategic syntheses for management, enabling a data-driven approach to AI ecosystem management.

6.2 Integration and orchestration mechanisms

The effectiveness of the AI Portal lies not only in the quality of single components but in the sophistication of integration mechanisms creating a coherent and functionally integrated user experience. These mechanisms operate on different levels to ensure navigation between quadrants and support areas is fluid, information is consistent and updated, and user can build work paths effectively combining diverse resources. First, the portal should implement an advanced semantic search system operating transversally across all quadrants and support areas, allowing users to rapidly identify use cases, documentary resources, prompts, codes, tools, and governance information relevant to a specific need or project. The search system should use natural language processing techniques to understand user intent and provide results not relying solely on textual matches but on semantic and contextual relevance. Second, every element present in the portal should be intelligently linked with related elements in other quadrants and areas through a dynamic link system based on relationship mapping defined in the framework. When a user explores a specific use case, the system should automatically present most relevant documentary resources, applicable prompts and codes, tools necessary for implementation, and pertinent governance considerations, creating guided discovery paths reducing cognitive complexity and accelerating time-to-value. The portal should then offer advanced personalization mechanisms allowing each user to configure their view according to functional needs, skills, and organizational responsibilities. These mechanisms should include possibility to create personalized dashboards aggregating most relevant use cases, configure alerts and notifications on updates of interest, maintain personal workspaces collecting projects, resources, and links to facilitate work continuity, and access personalized views of KPIs and metrics most relevant to their role. Considering the relevance (and sometimes confidentiality) of data used or produced, the AI Portal architecture should incorporate sophisticated governance mechanisms ensuring democratic access to AI resources is always aligned with corporate policies, compliance requirements, and security standards. These mechanisms should operate transparently for the user but provide AI Center of Excellence and governance structures with all tools necessary to maintain control and visibility over resource usage. In particular, access to each quadrant, support area, and specific resources within each section should be governed by an authorization system operating on different levels: organizational role, certified skills, assigned projects, information confidentiality level, and specific compliance requirements. This system should ensure each user has access to appropriate resources for their responsibilities while maintaining protection of sensitive information, tools requiring specialized skills, and functionalities that could have significant regulatory implications. All interactions with the portal should be systematically recorded to create a complete audit trail supporting governance, compliance, and optimization purposes. This logging system should record not only user accesses and actions but also usage patterns informing strategic decisions related to AI ecosystem evolution. The system should be designed specifically to support documentation and traceability requirements foreseen by AI Act and other relevant regulations, providing automated evidence of operational compliance. Awareness of investment value (to preserve and "monetize") the portal should incorporate user feedback systems allowing systematic collection of information on resource quality, tool effectiveness, gaps in available functionalities, and adequacy of governance policies and procedures. These mechanisms should feed continuous improvement processes ensuring the portal evolves in response to emerging organization needs, progress in available AI technologies, and regulatory context evolution.

6.3 Strategic implications and business value

The AI Portal represents much more than a technological interface: it constitutes the materialization of the organization's AI democratization and monetization strategy and the main vehicle for cultural transformation necessary for AI adoption success. Investment in realizing a sophisticated and well-designed portal generates business value on different dimensions going well beyond immediate operational efficiency. Availability of a unified and intuitive access point significantly reduces barriers to AI technology adoption, allowing users with different technical skill levels to effectively access and use resources that would otherwise remain underutilized. This democratization effect accelerates AI diffusion across the organization and increases overall value extracted from technological investments, while integrated governance mechanisms ensure this democratization occurs within controlled and compliant parameters. The portal, seen in this light, acts as a standardization mechanism ensuring AI usage across the organization is consistent with consolidated best practices, vendor instructions, corporate policies, and quality standards. This standardization reduces variability in obtained results, minimizes risks associated with improper technology use, and facilitates scalability of successful AI solutions across different departments and corporate functions. Systematic organization of all AI-related assets in a central and accessible repository constitutes a strategic investment in organizational knowledge management ensuring continuity and resilience regarding staff turnover, facilitating new collaborator onboarding, and preserving value of investments in skills and internally developed solutions. Systematic inclusion of vendor instructions and external best practices ensures critical knowledge remains not isolated but becomes accessible organizational asset. The portal should provide a platform for systematic collection of AI usage data informing strategic decisions regarding ecosystem evolution, resource allocation, and identification of new value opportunities. This measurement capability should transform AI management from a reactive approach to a strategic and data-driven one. Simultaneously, integrated governance mechanisms should ensure organization always maintains evidence of its regulatory compliance and can effectively respond to audit and control requirements. The AI Portal represents, ultimately, the operational synthesis of the organization's strategic vision regarding AI: a digital environment transforming AI ecosystem complexity into accessible opportunities, governing technological innovation through appropriate standards and controls, guaranteeing regulatory compliance in a complex regulatory context, and enabling cultural transformation necessary to realize full potential of artificial intelligence technologies in the contemporary corporate environment as can be an evolved system for managing other corporate intangibles (e.g., corporate performance and human resource management systems).